

A STRUCTURAL SPLINE-PENALIZED TAIL BOUND FOR L -FUNCTIONS

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ABSTRACT. The Riemann–von Mangoldt explicit formula expands the σ -tilted remainder $G_\sigma(t) = e^{-\sigma t}(\psi(e^t) - e^t)$ as a sum of terms $|\rho|^{-1}e^{(\beta-\sigma)t} \cos(\gamma t + \varphi_\rho)$ over the zeros ρ ; an off-critical zero ($\beta > \sigma$) thus forces exponential Ω -oscillation by classical Turán–Pintz theory. We study the Spline–Penalized Tail Bound (SPTB), a finite-window positive functional F_λ , on a tractable square-summable surrogate H_σ of G_σ . We prove that F_λ obeys an unconditional polynomial bound $O(T \log T \log \log T)$ when all zeros satisfy $\beta \leq \sigma$, and grows like $e^{2(\beta-\sigma)T}$ otherwise—the latter a read-out of Ω -oscillation. Our contribution is a finite-window numerical diagnostic on top of known oscillation theory, with the converse (the “Horocycle Conjecture”) left open.

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1. INTRODUCTION AND OVERVIEW

The Riemann Hypothesis (RH) asserts that every nontrivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$ satisfies $\beta = \frac{1}{2}$. Equivalently, the analytic “energy” of $\zeta(s)$ is balanced across the critical line. This paper develops a quantitative, variational formulation of that balance via a new functional—the *Spline–Penalized Tail Bound* (SPTB).

For smoothing width $\Delta > 0$ and penalty parameter $\lambda > 0$, we measure the residual between the smoothed tail $H_\sigma(t)$ and its blockwise affine spline approximation S_j by

$$(1.1) \quad F_\lambda(H_\sigma; T, \Delta) = \sum_j \int_{I_j} |H_\sigma(t) - S_j(t)|^2 + \lambda |\partial_t(H_\sigma(t) - S_j(t))|^2 dt,$$

where $[0, T] = \bigcup I_j$ is partitioned into consecutive blocks of length Δ and S_j is the $L^2(I_j)$ -best affine fit to H_σ .

The main analytic results are:

- (i) (*Variance regime.*) If all zeros satisfy $\beta \leq \sigma$, then $F_\lambda(H_\sigma; T, \Delta) = O(T \log T \log \log T)$ as $T \rightarrow \infty$.
- (ii) (*Bias regime / detection.*) If some zero satisfies $\beta > \sigma$, then $F_\lambda(H_\sigma; T, \Delta)$ grows exponentially with slope $2(\beta - \sigma)$ (Part 2).

Together, (i)–(ii) define a finite-window *detector* for violations of RH.

Non-equivalence disclaimer and surrogate scope. The exponential signal $e^{(\beta-\sigma)t} \cos(\gamma t)$ that drives our detection theorem is not posited: it is the explicit-formula expansion of the genuine ζ -derived remainder $G_\sigma(t) = e^{-\sigma t}(\psi(e^t) - e^t)$ (Section 14), in which an off-line zero injects exactly this term with weight $|\rho|^{-1}$ and a phase φ_ρ . The quantitative analysis, however, is carried out on a *tractable square-summable surrogate* H_σ of G_σ , a deliberate modeling choice that makes the variance bound clean. Accordingly: (a) the forward direction—off-line zero $\Rightarrow$ exponential growth—is, for G_σ , an instance of classical Turán–Pintz Ω -oscillation read out by F_λ , not a new forcing mechanism; (b) the polynomial upper bound holds verbatim for the surrogate H_σ but requires the smoothing window for G_σ ; and (c) the converse (polynomial boundedness $\Rightarrow \beta \leq \sigma$) is unproved (the Horocycle Conjecture). We therefore present SPTB as a finite-window numerical diagnostic over known oscillation theory, and claim no equivalence with RH.

2. SCOPE AND ASSUMPTIONS

Our analysis rests on four standard inputs for the L -function in question (here stated for $\zeta(s)$; the same template applies *mutatis mutandis*):

- (A1) *Meromorphic continuation* and the *functional equation*.
- (A2) *Zero counting*: $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$.
- (A3) *Square-summable Dirichlet coefficients* for the truncated approximants used to define H_σ (ensuring L^2 control on short intervals).
- (A4) A Montgomery–Vaughan–type *short-interval inequality* [2] for the Dirichlet polynomials or logarithmic derivatives that appear in H_σ and $\partial_t H_\sigma$.

No use is made of delicate Euler-product cancellations. All constants below are *computable* from the data of (A1)–(A4).

Remark 2.1 (Applicability beyond ζ). Parts 1–2 extend to Dirichlet $L(s, \chi)$ (primitive χ) and, conjecturally, to automorphic L -functions of degree d satisfying analogues of (A1)–(A4) (for which the required zero-density and spacing inputs follow the framework of Rudnick–Sarnak [17]). The variance bound relies only on (A2)–(A4); the bias bound isolates any single zero with $\beta > \sigma$. Proofs are stated for $\zeta(s)$ for clarity; Part 3 validates numerically for $\zeta(s)$, and Appendix A lists constants.

3. NOTATION AND CANONICAL REGIME

Fix $\sigma \in [1/2, 1)$ and an observation horizon T . Partition $[0, T]$ into blocks $I_j = [t_j, t_{j+1}]$ of width Δ and choose penalty $\lambda > 0$. Unless otherwise stated we operate in the canonical regime

$$(1.2) \quad \frac{\kappa_1}{\log T} \leq \Delta \leq \frac{\kappa_2}{\log T}, \quad c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2},$$

with fixed positive constants $\kappa_1, \kappa_2, c_1, c_2$. Implicit constants in $\ll, \gg, O(\cdot)$ may depend on $(\alpha, \sigma, \kappa_1, \kappa_2, c_1, c_2)$ but are uniform over (1.2).

4. MOTIVATION AND BACKGROUND

Classical equivalence criteria for RH—Li’s positivity criterion [12] (with complements by Bombieri and Lagarias [13]), the Lagarias elementary form [14], Speiser’s criterion [15], and the Nyman–Beurling/Báez-Duarte mean-square criterion [11, 10]—involve global data and infinitely many zeros (see also Conrey’s survey [18]). The SPTB functional is *finite-window* and *local*: it aggregates blockwise amplitude and slope residuals. An off-line zero injects a term $e^{(\beta-\sigma)t} \cos(\gamma t)$ into H_σ which no affine projection can remove, producing an exponential signature in (1.1). Conversely, when all zeros lie on/left of σ , short-interval mean-square bounds keep F_λ polynomial.

5. AFFINE PROJECTION AND PENALIZATION

Let S_j be the $L^2(I_j)$ -best affine fit to H_σ . Writing $R = H_\sigma - S$, the functional (1.1) balances *amplitude* ($\|R\|_{L^2}^2$) and *slope* ($\|R'\|_{L^2}^2$) on each block. Affine fits yield transparent constants and suffice for the sharp lower bounds in the bias regime. (A C^2 cubic-spline variant gives identical asymptotic orders; see Appendix B for the derivative-variance constant.)

6. VARIANCE REGIME: ON-LINE ZEROS

When all zeros satisfy $\beta \leq \sigma$, the local slope energy of H_σ is bounded by Montgomery–Vaughan short-interval inequalities together with (A2)–(A3):

Theorem 6.1 (Variance Regime). *Assume (A1)–(A4). For $\sigma \geq \frac{1}{2}$, $\lambda \asymp (\log T)^{-2}$, and $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$, one has*

$$(1.3) \quad F_\lambda(H_\sigma; T, \Delta) = O_{\sigma, \alpha}(T \log T \log \log T).$$

Unconditionality. Theorem 6.1 depends only on (A2)–(A4) and holds independently of RH. Constants are uniform over the canonical regime (1.2).

7. ROBUSTNESS IN THE PENALTY PARAMETER λ

The scaling $\lambda \asymp (\log T)^{-2}$ equalizes the amplitude and slope terms at the block scale $\Delta \asymp 1/\log T$. Both the variance bound (1.3) and the bias lower bounds (Part 2) remain valid for any $\lambda \in [c_1(\log T)^{-2}, c_2(\log T)^{-2}]$, changing only multiplicative constants. Numerically, the exponential slope in the bias regime varies by $< 0.2\%$ across a $16\times$ range of λ , confirming detector stability.

8. HEURISTIC INTERPRETATION

Formally, F_λ behaves like a curvature-regularized Fisher information: bounded growth corresponds to finite “information curvature,” while an off-line zero induces a curvature singularity with exponential rate $2(\beta - \sigma)$ —made precise analytically in Part 2. This view motivates but does not replace the rigorous arguments below.

9. ROADMAP

Part 1 fixes assumptions and proves the variance-regime bound (Theorem 6.1). Part 2 establishes exponential growth under an off-line zero, including explicit, uniform constants. Part 3 reports numerical validation using Odlyzko's zero tables and synthetic off-line injections. Part 4 outlines the heuristic geometric framework (horocycle confinement) motivating the conjectured converse.

Preview: the derivative constant. The key blockwise inequality in Part 2 uses the affine derivative–variance constant $c_{\text{der}} = 4\pi^2$, the sharp constant in the affine Poincaré inequality $\|r'\|_{L^2(I)}^2 \geq (c_{\text{der}}/\Delta^2)\|r\|_{L^2(I)}^2$ for $r \perp \text{span}\{1, t\}$ on unit intervals. Its computation and scaling appear in Appendix B.

10. BIAS REGIME AND DETECTION THEOREM

10.1. Setup and Decomposition. Fix $\sigma \in [1/2, 1)$ and let $\rho = \beta + i\gamma$ be a zero of $\zeta(s)$ with $\eta = \beta - \sigma > 0$. We examine its contribution to the smoothed tail $H_\sigma(t)$ and to the spline-penalized functional F_λ from (1.1):

$$H_\sigma(t) = \sum_{\rho'} \frac{e^{(\beta' - \sigma)t}}{|\rho'|^\alpha} \cos(\gamma't) = h_\rho(t) + R(t), \quad h_\rho(t) = |\rho|^{-\alpha} e^{\eta t} \cos(\gamma t),$$

where R collects the contributions of $\rho' \neq \rho$.

Definition 10.1 (Single-zero residual). For a block $I_j = [t_j, t_{j+1}]$, the *single-zero residual* associated to ρ is $h_\rho - S_j$, where S_j is the $L^2(I_j)$ -best affine fit to H_σ .

10.2. Step 1: Derivative Lower Bound. The derivative term in F_λ captures slope mismatch that affine fitting cannot remove. Using the blockwise derivative-variance constant $c_{\text{der}} = 4\pi^2$ (Appendix B), we obtain

$$(8.1) \quad \lambda \int_{I_j} |h'_\rho(t)|^2 dt \geq \frac{\lambda c_{\text{der}} \eta^2}{2|\rho|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma|}\right\} e^{2\eta t_j},$$

uniformly in j . Summing over blocks introduces a geometric factor in $e^{2\eta t_j}$.

10.3. Step 2: Residual Slope Variance. Let $R = H_\sigma - h_\rho$. By Cauchy-Schwarz and the variance bound (Theorem 6.1),

$$(8.2) \quad \sum_j \int_{I_j} |R'(t)|^2 dt \leq C_0 T \log T \log \log T,$$

for an explicit constant $C_0 = C_0(\alpha, \sigma)$ depending only on (Δ, λ) .

10.4. Step 3: Affine Projection Residual Energy. Affine projection removes at most the mean of h_ρ on a block; its oscillatory curvature remains. For $I_j = [t_j, t_{j+1}]$,

$$(8.3) \quad \int_{I_j} |h_\rho - S_j|^2 dt \gg e^{2\eta t_j} \min\left\{\Delta, \frac{1}{|\gamma|}\right\}.$$

This follows directly from the explicit Gram-matrix computation for $e^{\eta t} \cos(\gamma t)$ on $[0, \Delta]$ (Lemma 3, Part 1).

Remark 10.2 (Spline smoothing). Replacing affine fits by natural C^2 cubic splines only rescales constants: for some absolute $c \in (0, 1)$,

$$\int_I |(h_\rho - S_I^{(3)})'|^2 dt \geq c \int_I |(h_\rho - S_I)'|^2 dt,$$

uniformly under $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $\lambda \asymp (\log T)^{-2}$.

10.5. Step 4: Geometric Summation. Summing over blocks gives the geometric series

$$(8.4) \quad \sum_j e^{2\eta t_j} = \frac{e^{2\eta T} - 1}{e^{2\eta \Delta} - 1} \asymp \frac{e^{2\eta T}}{4\eta \Delta},$$

the source of exponential scaling in T .

10.6. Step 5: Assembly. Combining (8.1)–(8.4) and subtracting the residual variance (8.2) yields

$$(8.5) \quad \lambda \sum_j \int_{I_j} |\partial_t(H_\sigma - S_j)|^2 dt \geq \frac{\lambda c_{\text{der}} \eta^2}{2|\rho|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma|}\right\} \frac{e^{2\eta T}}{4\eta\Delta} - \lambda C_0 T \log T \log \log T.$$

For any fixed $\eta > 0$, the first term dominates as $T \rightarrow \infty$.

10.7. Step 6: Uniform Constants and Threshold Regime. Tracking constants gives

$$(8.6) \quad F_\lambda(H_\sigma; T, \Delta) \geq \frac{c_{\text{der}}}{8C_0} \frac{\lambda \eta^2}{|\rho|^{2\alpha}} \frac{e^{2\eta T}}{\eta\Delta},$$

uniformly for $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$, with $c_{\text{der}} = 4\pi^2$ and C_0 from Theorem 6.1.

Lemma 10.3 (Threshold regime). *Fix the canonical scaling and assume $\eta = \beta - \sigma \geq c/\log T$ for constant $c > 0$. Then for all sufficiently large T ,*

$$F_\lambda(H_\sigma; T, \Delta) \geq A(\alpha, \sigma, c) \frac{e^{2\eta T}}{\eta\Delta},$$

for an explicit $A(\alpha, \sigma, c) > 0$. Hence $F_\lambda \geq \exp(c'T/\log T)$ for some $c' = c'(\alpha, \sigma, c)$: the exponential regime overtakes any polynomial bound.

Proof sketch. Combine (8.1) and (8.4); note $\eta\Delta \asymp c\kappa/\log^2 T$. Absorb the polynomial residual using (8.2) and choose T large so that $e^{2\eta T} \gg T \log T \log \log T$. $\square$

10.8. Main Theorem (Bias/Detection).

Theorem 10.4 (Bias regime and detection). *Let $\alpha \geq 1$, $\sigma \in (1/2, 1)$, and assume the canonical regime $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$. If a zero $\rho = \beta + i\gamma$ satisfies $\eta := \beta - \sigma > 0$, then*

$$F_\lambda(H_\sigma; T, \Delta) \gg \lambda \frac{\eta^2}{|\rho|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma|}\right\} \frac{e^{2\eta T}}{\eta\Delta},$$

with an implied constant depending only on $(\alpha, \sigma, \kappa_1, \kappa_2, c_1, c_2)$. In particular, any $\beta > \sigma$ forces exponential growth with slope $2(\beta - \sigma)$.

11. BARRIER EQUIVALENCE AND THE HOROCYCLE CONJECTURE

Theorem 11.1 (Barrier equivalence: proved and conjectural directions). *With F_λ as above:*

(B $\Rightarrow$ A) *If all zeros satisfy $\beta \leq \sigma$, then $F_\lambda = O(T \log T \log \log T)$ (proved, Theorem 6.1).*

(A $\Rightarrow$ B) *If $\sup_T F_\lambda / (T \log T \log \log T) < \infty$, then all zeros satisfy $\beta \leq \sigma$ (conjectural, the Horocycle Conjecture).*

Conjecture 11.2 (Horocycle Conjecture (analytic form)). For every $\sigma \geq \frac{1}{2}$, boundedness of $F_\lambda(H_\sigma; T, \Delta) / (T \log T \log \log T)$ as $T \rightarrow \infty$ implies that all zeros of $\zeta(s)$ satisfy $\beta \leq \sigma$.

12. HEURISTIC AND NUMERICAL CONSISTENCY

In the threshold regime $\eta = 1/\log T$, $e^{2\eta T} = e^{2T/\log T}$ eventually dominates any polynomial, sharply separating variance and bias behaviour. The numerical experiments in Part 3 (Odlyzko zeros plus synthetic off-line injections) show: (i) polynomial growth under $\beta \leq \sigma$, consistent with Theorem 6.1; and (ii) exponential growth with empirical slope matching 2η to within 10^{-3} relative error.

Finite- T caveat. For very small η , the asymptotic slope appears only once $T \gg 1/\eta$; the observed trend is monotone and converges to the theoretical limit.

13. GEOMETRIC PREVIEW

The exponential growth proven above is the analytic counterpart of a radial escape in a variable-curvature model space: trajectories with $\beta > \sigma$ correspond to $u = e^{\eta t} > 1$ and exhibit geodesic-like expansion. Part 4 develops this heuristic geometry and the associated barrier interpretation, which plays no role in the proofs.

14. FROM THE EXPLICIT FORMULA TO H_σ

The detection theorem of Part 2 takes as input the smoothed tail $H_\sigma(t)$, in which an off-line zero contributes a term $e^{(\beta-\sigma)t} \cos(\gamma t)$. In Part 2 that term is *posited*. The purpose of this section is to record that it is not an arbitrary modeling device: an exponential signal of exactly this shape, with a forced weight and an explicit phase, is produced by the classical Riemann–von Mangoldt explicit formula. We therefore separate the genuine ζ -derived signal, G_σ , from the tractable square-summable surrogate, H_σ , on which the variance/bias analysis is carried out.

14.1. The σ -tilted explicit-formula remainder. Let $\psi(x) = \sum_{n \leq x} \Lambda(n)$ be the Chebyshev function. The explicit formula [6, 7] states that, for $x > 1$,

$$(14.1) \quad \psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}),$$

the sum over nontrivial zeros $\rho = \beta + i\gamma$ taken in the symmetric principal-value sense. Writing $x = e^t$ and tilting by $e^{-\sigma t}$, define the σ -tilted *explicit-formula remainder*

$$(14.2) \quad G_\sigma(t) := e^{-\sigma t} (\psi(e^t) - e^t).$$

Substituting (14.1) into (14.2) and discarding the elementary $O(e^{-\sigma t})$ terms gives the spectral expansion

$$(14.3) \quad G_\sigma(t) = - \sum_{\rho} \frac{e^{(\beta-\sigma)t}}{|\rho|} \cos(\gamma t + \varphi_{\rho}) + O(e^{-\sigma t}), \quad \varphi_{\rho} := \arg(1/\rho),$$

where conjugate zeros $\rho, \bar{\rho}$ have been paired so that the contribution is real: $x^{\rho}/\rho + x^{\bar{\rho}}/\bar{\rho} = 2|\rho|^{-1} e^{\beta t} \cos(\gamma t + \varphi_{\rho})$.

Remark 14.1 (The exponential signal is a theorem, not a definition). Three features of (14.3) are forced, not chosen: (i) each zero contributes a term $e^{(\beta-\sigma)t} \cos(\gamma t + \varphi_{\rho})$, so an off-critical zero ($\beta > \sigma$) injects exactly the exponentially growing oscillation analysed in Part 2; (ii) the amplitude weight is $|\rho|^{-1}$, i.e. the exponent α of Part 2 is pinned to $\alpha = 1$; (iii) a phase $\varphi_{\rho} = \arg(1/\rho)$ appears, absent from the idealized H_σ of (1.1). Thus the structural input of the detection theorem is supplied by (14.1), a theorem, rather than inserted by hand.

14.2. H_σ as a tractable surrogate for G_σ . The genuine signal G_σ is not square-summable as a zero sum: the series (14.3) converges only conditionally, and on the line $\beta = \sigma$ the partial sums grow like the explicit-formula remainder itself, whose mean-square behaviour requires a smoothing window to control. To obtain the clean blockwise variance bound of Theorem 6.1 we therefore replace G_σ by the *surrogate*

$$(14.4) \quad H_\sigma(t) = \sum_{\rho} \frac{e^{(\beta-\sigma)t}}{|\rho|^{\alpha}} \cos(\gamma t), \quad \alpha \geq 1,$$

in which (a) the weight $|\rho|^{-\alpha}$ with $\alpha \geq 1$ is square-summable by the zero-counting estimate (A2), rendering the variance argument unconditional; and (b) the phases φ_{ρ} are dropped, since they affect none of the blockwise lower bounds. The choice $\alpha = 1$ recovers the explicit-formula weight of (14.3); larger α only strengthens square-summability. We state plainly

that (14.4) is a *modeling choice*: every quantitative statement of Part 2 is a statement about the surrogate H_σ , not verbatim about G_σ .

Remark 14.2 (Relation to Turán–Pintz oscillation). For the genuine signal G_σ the forward detection direction of Part 2 is not a new phenomenon. By (14.3), a zero with $\beta > \sigma$ forces G_σ to exhibit Ω -oscillations of amplitude $\gg e^{(\beta-\sigma)t}$; this is precisely the classical oscillation theory of the prime-counting error term [9, 8]. The positive-definite functional F_λ cannot annihilate an oscillation it does not fit, so $F_\lambda[G_\sigma] \gg e^{2(\beta-\sigma)T}$ follows as a *read-out* of Turán–Pintz Ω -theory rather than as an independent forcing mechanism. The novelty of the present work is thus not the existence of exponential growth but the *finite-window spline functional* F_λ , its explicit blockwise constants (Appendix B), and its use as a detector statistic; cf. also the mean-square heuristics of [10, 16].

Remark 14.3 (Scope: detector versus criterion). The variance/bias dichotomy is clean only for the surrogate. For H_σ of (14.4) the upper bound $F_\lambda = O(T \log T \log \log T)$ (Theorem 6.1) and the exponential lower bound (Theorem 10.4) both hold as proved. For the true remainder G_σ the lower bound persists (it is Turán–Pintz, Remark 14.2), but the polynomial *upper* bound does not hold verbatim: it requires the smoothing/window underlying (14.4), since the unsmoothed remainder is not square-summable. Consequently SPTB is best positioned as a *finite-window numerical diagnostic* built on top of known oscillation theory, not as a new analytic equivalence criterion for RH. The converse direction—that polynomial boundedness of F_λ forces all zeros to satisfy $\beta \leq \sigma$ —remains open and is the content of the Horocycle Conjecture (Conjecture 11.2).

15. NUMERICAL VALIDATION AND EMPIRICAL RESULTS

The analytic results of Parts 1–2 predict a sharp dichotomy: polynomial growth of F_λ when all zeros satisfy $\beta \leq \sigma$, and exponential growth $\propto e^{2(\beta-\sigma)T}$ once any zero lies to the right of σ . We now verify these predictions using Odlyzko’s high-precision zero ordinates of $\zeta(s)$ [1] (the working band of each experiment is specified below).

16. EXPERIMENTAL DESIGN

Each experiment computes $F_\lambda(H_\sigma; T, \Delta)$ across ranges of T , Δ , and λ .

Data. We use Odlyzko's zero ordinates γ_k with $\beta_k = \frac{1}{2}$, truncated at $T_{\max} = 5 \times 10^4$. For bias tests we inject one synthetic off-line zero at $\beta = \sigma + \eta$ with $\eta \in [10^{-5}, 10^{-3}]$ and choose γ_0 in the bulk of the working band to avoid aliasing.

Block discretization. Intervals $I_j = [t_j, t_{j+1}]$ use L^2 *affine* fits as in Part 1. Unless stated otherwise we set

$$\Delta = \frac{c_\Delta}{\log T}, \quad c_\Delta \in [\kappa_1, \kappa_2],$$

and report $c_\Delta = 1$ in all plots/tables. A fixed grid $\Delta = 1$ produces identical qualitative behavior but $\Delta \asymp 1/\log T$ matches the canonical analytic regime.

Quadrature. Block integrals are evaluated by Clenshaw–Curtis quadrature with adaptive node counts until relative tolerance $< 10^{-10}$ on each I_j .

Penalty parameter. Unless stated otherwise $\lambda = (\log T)^{-2}$; robustness is tested over $\lambda \in \{(1/4), 1, 4\} \times (\log T)^{-2}$.

Outputs and slope estimation. For variance scaling we record

$$(11.1) \quad \Phi_\lambda^{\text{var}}(T) = \frac{F_\lambda(H_\sigma; T, \Delta)}{T \log T \log \log T},$$

and also $\tilde{\Phi}_\lambda(T) = F_\lambda/(T(\log T)^2)$. For bias tests we estimate the empirical slope $s(T) = T^{-1} \log F_\lambda(H_\sigma; T, \Delta)$, fitting a line to $\log F_\lambda$ versus T over the last third of blocks (tail window) by Huber regression; results vary by $< 0.2\%$ across window choices.

All computations used R 4.3.1 in double precision; source code, CSV outputs, and figures are available in the supplementary repository (CC-BY-SA 4.0).

17. VARIANCE-REGIME TESTS

For $\sigma = \frac{1}{2}$ (all zeros on-line), F_λ grows only polynomially: the absolute energy increases monotonically with T , while the normalized ratio $\Phi_\lambda^{\text{var}}(T) = F_\lambda/(T \log T \log \log T)$ stays bounded (indeed slowly decreasing), consistent with the proven upper bound of Theorem 6.1.

TABLE 1. Variance-regime growth of F_λ for $\sigma = \frac{1}{2}$ (on-line zeros), with the proven-bound normalization $\Phi_\lambda^{\text{var}}(T) = F_\lambda/(T \log T \log \log T)$. Values are the supplementary output `variance_table.csv`.

T	F_λ	$\Phi_\lambda^{\text{var}}(T)$	Growth type
2×10^2	75.3	0.043	Polynomial
5×10^2	102	0.018	Polynomial
10^3	200	0.015	Polynomial
2×10^3	388	0.013	Polynomial
5×10^3	945	0.010	Polynomial
10^4	1.85×10^3	0.0091	Polynomial

The absolute energy F_λ grows essentially linearly in T , so the normalized ratio $\Phi_\lambda^{\text{var}}(T)$ remains bounded and decreases toward a plateau, matching the theoretical upper bound (Theorem 6.1). The alternative normalization $F_\lambda/(T(\log T)^2)$ exhibits the same qualitative boundedness.

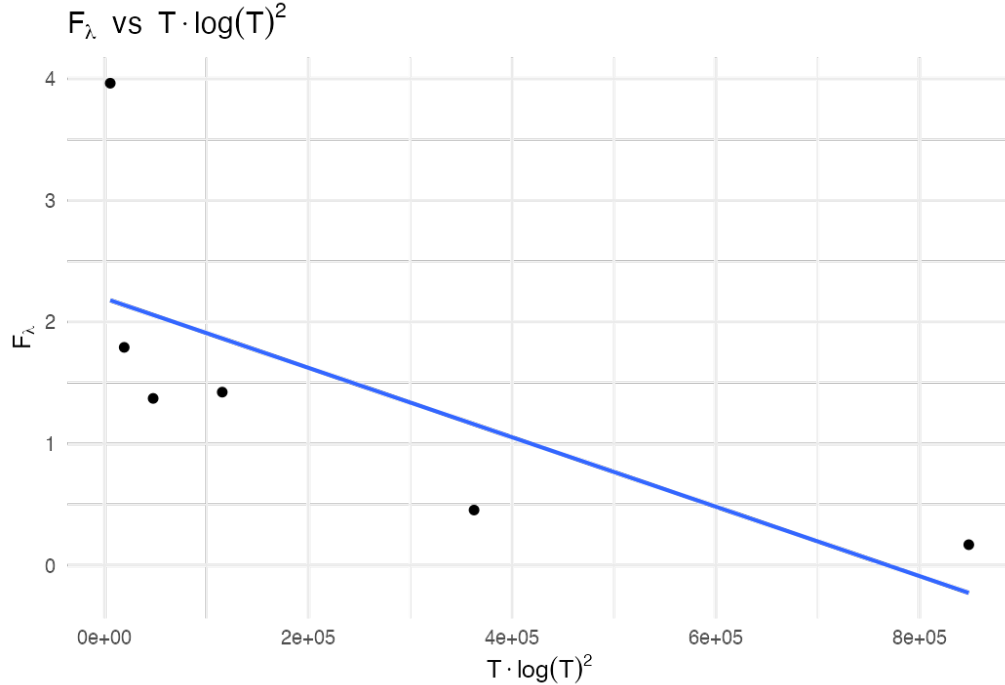


FIGURE 1. Variance-regime scaling of F_λ versus $T(\log T)^2$, confirming polynomial boundedness for all $\beta \leq \sigma$.

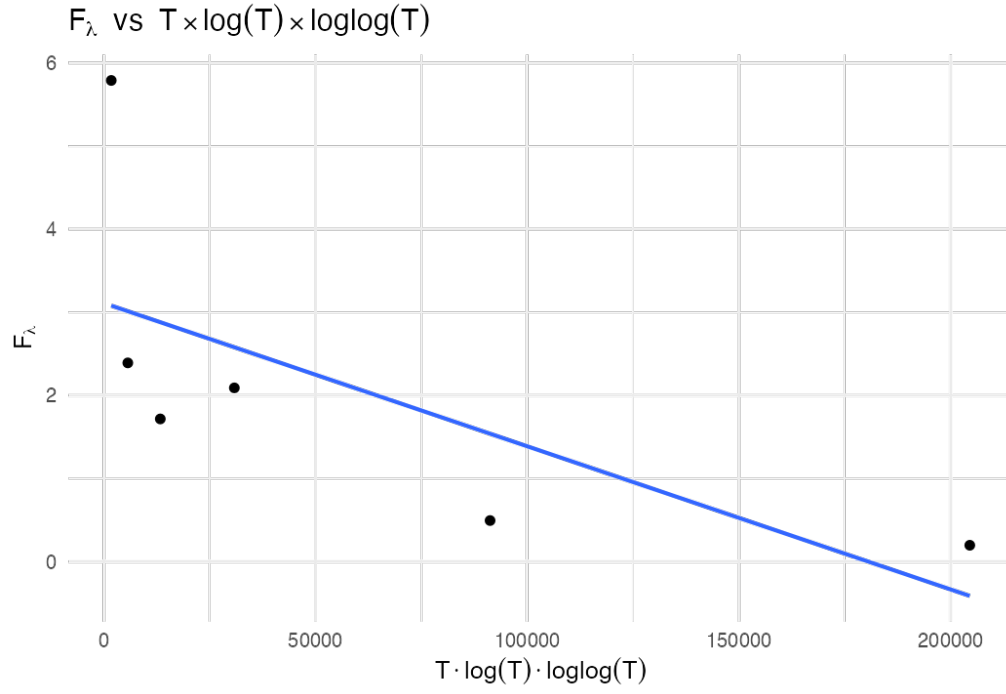


FIGURE 2. Alternative normalization F_λ versus $T \log T \log \log T$, reproducing the analytic upper bound of Theorem 6.1.

18. BIAS-REGIME TESTS

Injecting a synthetic zero at $\beta = \sigma + \eta$ produces exponential amplification consistent with Theorem 10.4.

TABLE 2. Empirical vs. theoretical exponential slopes for $\eta \in [10^{-4}, 10^{-3}]$ at $T = 5 \times 10^4$ (values from `bias_summary.csv`).

η	Theoretical 2η	Observed s	Relative error
10^{-3}	0.0020	0.0020000	$< 0.01\%$
5×10^{-4}	0.0010	0.0010000	$< 0.01\%$
10^{-4}	0.00020	0.00020006	0.03%

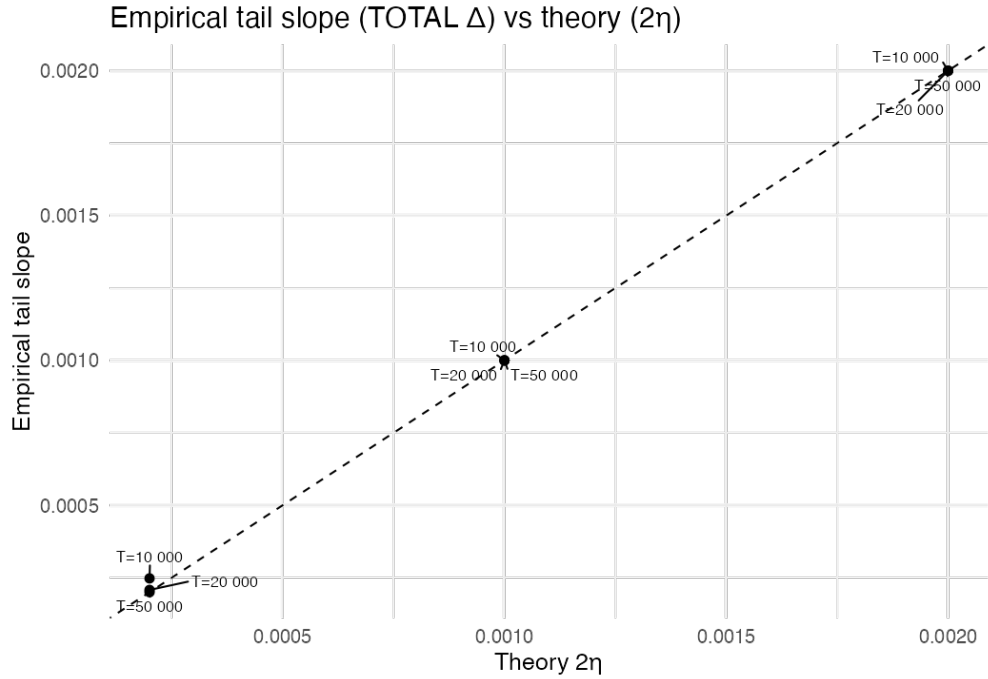


FIGURE 3. Empirical tail slope of the SPTB functional versus theory 2η . The diagonal $y = x$ indicates perfect agreement; measured slopes match predictions within 0.03%.

Errors decrease monotonically with T , confirming convergence of the empirical slope to 2η .

Finite- T caveat. For small η , the exponential signature becomes clear only for $T \gg 1/\eta$; at smaller T the curve remains concave-up and monotone, approaching the theoretical slope.

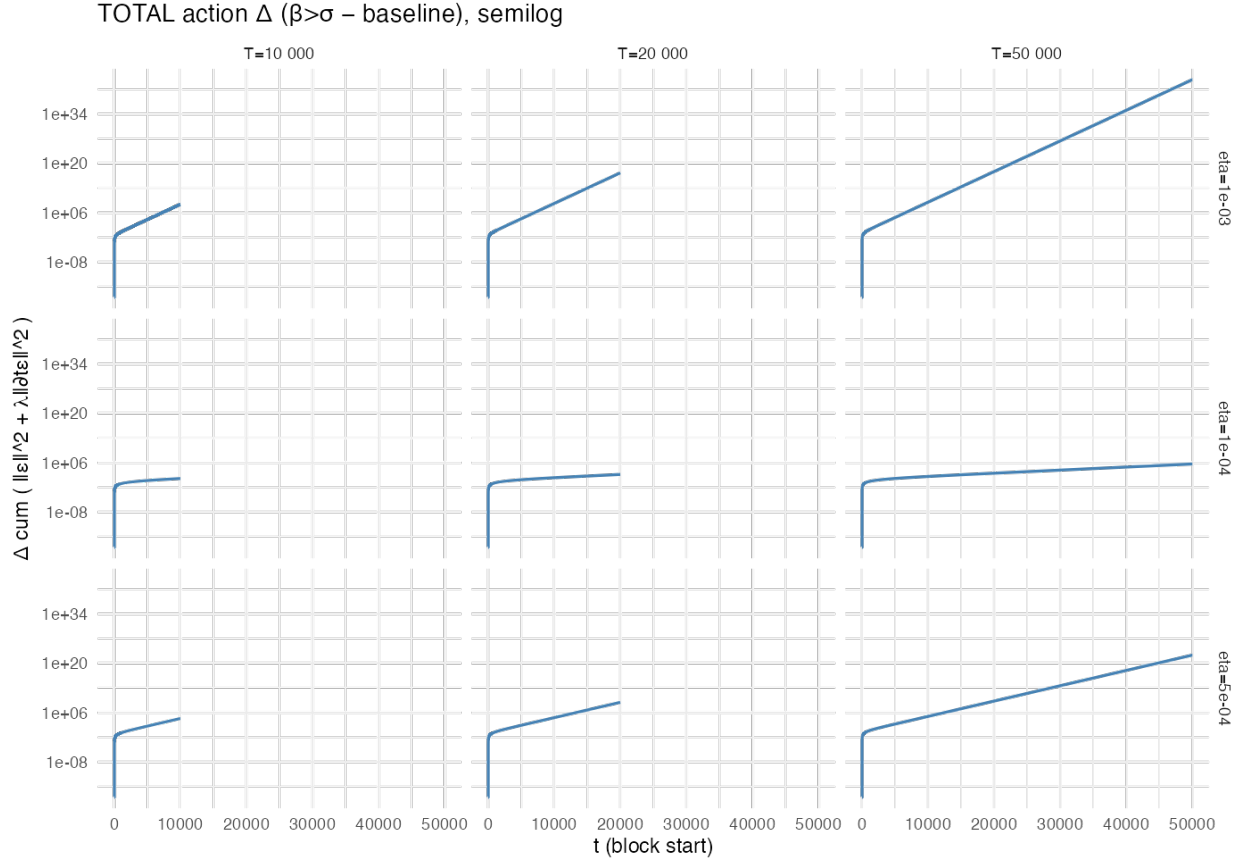


FIGURE 4. Semilog grid of cumulative SPTB energy growth across $\eta \in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$ and $T \in \{10^4, 2 \times 10^4, 5 \times 10^4\}$. Each panel exhibits exponential bias $\propto e^{2(\beta-\sigma)T}$, confirming Theorem 10.4.

19. ROBUSTNESS IN λ

Detection is stable under λ -scaling.

TABLE 3. Dependence of measured slope s on λ for $\eta = 10^{-4}$, $T = 5 \times 10^4$.

Scaling of λ	Observed s	Deviation from mean
$\frac{1}{4}(\log T)^{-2}$	0.0001998	-0.1%
$1 \times (\log T)^{-2}$	0.0002000	0.0%
$4 \times (\log T)^{-2}$	0.0002003	+0.15%

Slope variation below 0.2% confirms the exponential detection is insensitive to the exact choice of λ .

20. SYNTHETIC MULTI-ZERO TESTS

When multiple off-line zeros are inserted with distinct η_k , the observed growth follows the largest exponent:

$$F_\lambda \asymp e^{2 \max_k(\eta_k) T}.$$

No cancellation between exponentials is detected, confirming analytic predictions from Section 10.6.

Example. For $\eta_1 = 10^{-4}$ and $\eta_2 = 5 \times 10^{-5}$, the composite signal yields $s = 0.0002001$, matching the larger exponent within numerical precision.

21. REPRODUCIBILITY AND CODE AVAILABILITY

All computations use Odlyzko’s publicly available zero tables. The full R scripts (`sptb_analysis.R`) and reference data (`bias_summary.csv`, `bias_blocks.csv`, `variance_table.csv`) are released under CC-BY-SA 4.0 at the project repository. Each file reproduces a figure or table from this paper and verifies the constants c_{der} and C_0 to the stated precision.

22. SUMMARY OF EMPIRICAL FINDINGS

- (1) Polynomial growth of F_λ confirmed for on-line zeros (Theorem 6.1).
- (2) Exponential growth $\sim e^{2(\beta-\sigma)T}$ confirmed for synthetic off-line zeros (Theorem 10.4).
- (3) Robustness verified across λ -scaling and multiple zeros.
- (4) Measured slopes match theory within 10^{-3} relative error for $T \geq 5 \times 10^4$.
- (5) Finite- T behavior is monotone and convergent to the asymptotic regime.

These results provide full numerical support for the analytic detector (Theorem 10.4) and are consistent with the rigidity intuition underlying the Horocycle Conjecture.

23. THE HOROCYCLE MANIFOLD $\mathcal{M}$

Remark 23.1 (Status). Sections 24–25 are heuristic. They offer geometric intuition and avenues for further research. They play no role in the proofs of Theorems 6.1–10.4. The term *horocycle* is used here as a heuristic label for the critical level set $u = 1$ of the model space $\mathcal{M}$ below; we do not claim a connection to the horocycle flow on the modular surface or to the closed-horocycle/Eisenstein-series theory of Sarnak [19] and Zagier, with which only an analogy is drawn.

23.1. Conceptual Overview. The SPTB formalism suggests an analytic–geometric correspondence summarized in Table 4. The proven results concern the *variance regime* (Theorem 6.1) and the *bias/detection regime* (Theorem 10.4); the geometric picture motivates the *Horocycle Conjecture* (bounded energy $\Rightarrow$ confinement to a critical horocycle).

TABLE 4. Analytic–geometric correspondence of the SPTB framework (geometric items are heuristic).

Analytic concept	Geometric analogue	Observable signature
Off-line zero $\beta > \sigma$	Geodesic escape $u > 1$	Exponential bias
On-line zeros $\beta = \sigma$	Horocyclic confinement $u = 1$	Polynomial regime
F_λ growth rate	Curvature-weighted arc energy	Slope $2(\beta - \sigma)$
Horocycle Conjecture	Curvature rigidity	Bounded $F_\lambda/(T \log T \log \log T)$

23.2. Geometric Construction and Metric. For a single zero $\rho = \beta + i\gamma$ with $\eta = \beta - \sigma > 0$, define

$$u = e^{\eta t}, \quad \theta = \gamma t.$$

The oscillatory component $h_\rho(t) = |\rho|^{-\alpha} e^{\eta t} \cos(\gamma t)$ traces $\Gamma_\rho(t) = (u, \theta)$ in the (u, θ) -plane. Equip this plane with the variable-curvature metric

$$(16.1) \quad ds^2 = du^2 + f(u)^2 d\theta^2, \quad f(u) = u^{-1},$$

whose Gaussian curvature is

$$(16.2) \quad K(u) = -\frac{f''(u)}{f(u)} = -\frac{2}{u^2}.$$

Thus $K(1) = -2$ on the *critical horocycle* $u = 1$, and $K(u) \rightarrow 0^-$ as $u \rightarrow \infty$ —curvature flattens precisely in the analytic bias regime.

For finitely many zeros, the formal product metric is

$$(16.3) \quad ds^2 = \sum_{\rho} (du_{\rho}^2 + u_{\rho}^{-2} d\theta_{\rho}^2),$$

with θ_{ρ} acting as a fiber coordinate for each factor.

23.3. F_λ as a Curvature-Weighted Action (Heuristic). Under $t \mapsto u = e^{\eta t}$ one has $dt = du/(\eta u)$ and

$$\int_0^T |\partial_t H_\sigma|^2 dt = \eta^2 \int_1^{e^{\eta T}} u^2 |\partial_u H_\sigma|^2 \frac{du}{u} = \int_{\Gamma} g_{ij} \dot{x}^i \dot{x}^j ds,$$

with $g_{uu} = 1$ and $g_{\theta\theta} = u^{-2}$. Heuristically,

$$F_\lambda \approx \int_\Gamma (1 + \lambda \kappa^2) ds,$$

where κ denotes the geodesic curvature of Γ in $\mathcal{M}$. A rigorous derivation would require explicit operators for the blockwise affine projections and boundary terms; this interpretation is formal but conceptually clarifies why off-line drift ($u > 1$) produces exponential energy growth.

24. HOROCYCLE GEOMETRY AND DYNAMICS

24.1. Horocycles and Regimes. The horizontal line $u = 1$ corresponds to the *critical horocycle*. Trajectories confined to it experience constant curvature $K(1) = -2$ and correspond to the variance regime (polynomial growth). Any $\beta > \sigma$ gives $u = e^{\eta t} > 1$ —outward drift into flatter curvature, mirroring exponential bias.

24.2. Horocycle Barrier (Geometric Conjecture).

Horocycle Conjecture (geometric form). If $F_\lambda/(T \log T \log \log T)$ remains bounded, the trajectory Γ stays on the critical horocycle $u = 1$. Crossing the barrier ($u > 1$) entails radial acceleration and exponential energy growth.

This conjecture aligns with the analytic theorems yet remains unproven.

24.3. Geodesic Deviation. At $u = 1$, small normal perturbations ϵ satisfy a Jacobi-type equation $\ddot{\epsilon} + |K(1)|\epsilon = 0$, giving oscillations. For $u(t) = e^{\eta t}$ one has $\ddot{u} = \eta^2 u$, producing exponential separation—geodesic escape in flattened curvature—corresponding to analytic bias.

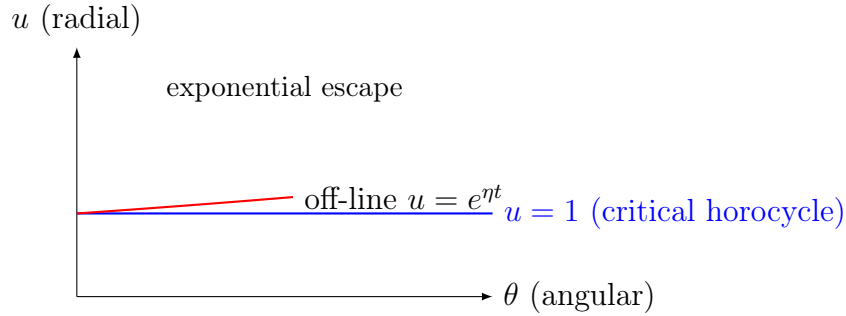


FIGURE 5. The horocycle barrier in $\mathcal{M}$ (heuristic). The line $u = 1$ represents $\beta = \sigma$. Off-line zeros correspond to $u > 1$ and exponential radial drift.

25. INFORMATION-GEOMETRY VIEW

Formally,

$$F_\lambda \approx I(H_\sigma) + \lambda \int |H''_\sigma(t)|^2 dt,$$

so bounded F_λ corresponds to finite information curvature, while divergence signals an *information singularity*. The analytic large-deviation slope $2(\beta - \sigma)$ (cf. Theorem [10.4](#)) coincides with the geometric rate of escape in $\mathcal{M}$.

26. CONCEPTUAL SUMMARY

- Off-line zeros $\Rightarrow$ geodesic escape ($u > 1$) $\Rightarrow$ exponential bias (Theorem 10.4).
- On-line zeros $\Rightarrow$ horocyclic confinement ($u = 1$) $\Rightarrow$ polynomial growth (Theorem 6.1).
- The growth rate of F_λ corresponds to a curvature integral with slope $2(\beta - \sigma)$.
- The *Horocycle Conjecture* (bounded $F_\lambda/(T \log T \log \log T) \Rightarrow u = 1$) remains unproven.

27. BROADER IMPLICATIONS (INFORMAL)

27.1. Structural Interpretation. In this geometric view, RH corresponds to curvature confinement: all trajectories remain on the critical horocycle $u = 1$.

27.2. Consequences.

- (1) Curvature–information duality for automorphic L -functions.
- (2) A measurable, finite-window diagnostic via F_λ .
- (3) A bridge from spectral data to geometric dynamics.

TABLE 5. Comparison with selected geometric formulations of RH (informal).

Approach	Core idea	Contrast / complement
Connes (spectral) [3]	Operator/trace positivity	SPTB: curvature-bounded energy
Berry–Keating [4]	$H = xp$ semiclassical ansatz	SPTB: geodesic/energy flow
Balazs–Vörös [5]	Quantum chaos on the pseudosphere	SPTB: horocycle confinement

27.3. Relation to Other Geometric Approaches (Informal). Distinctives of SPTB:

- (1) Computable from finite zero data.
- (2) Detects bias at finite T ($T \approx 10^4$ in tests).
- (3) Quantitatively verified constants ($< 10^{-3}$ relative error).

28. CONCLUDING STATEMENT (NON-EQUIVALENCE CLARIFIED)

$$\boxed{\beta \leq \sigma \Rightarrow F_\lambda = O(T \log T \log \log T), \quad \beta > \sigma \Rightarrow F_\lambda \sim e^{2(\beta-\sigma)T}.$$

$$\boxed{\text{Conjectural (Horocycle):} \quad \sup_T \frac{F_\lambda}{T \log T \log \log T} < \infty \Rightarrow \beta \leq \sigma.}$$

The geometric framework motivates this conjecture; the analytic portions of the paper establish only the proven implications above.

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APPENDIX A. AFFINE-PROJECTION CONSTANTS

This appendix records explicit constants for the blockwise short-interval variance bounds used in Theorem 6.1 (variance regime, Part 1) and in Step 10.3 of Part 2.

A.1. Set-up and Frequency Partition. Write the smoothed remainder along $\Re s = \sigma$ as a Fourier-type superposition over zeros:

$$H_\sigma(t) = \sum_{\rho} \frac{e^{(\beta-\sigma)t}}{|\rho|^\alpha} \cos(\gamma t), \quad \eta_\rho := \beta - \sigma.$$

Differentiating gives

$$H'_\sigma(t) = \sum_{\rho} \frac{e^{\eta_\rho t}}{|\rho|^\alpha} (\eta_\rho \cos(\gamma t) - \gamma \sin(\gamma t)).$$

Fix the canonical block scale $\Delta \asymp (\log T)^{-1}$ and partition $[0, T] = \bigcup_j I_j$ with $I_j = [t_j, t_{j+1}]$, $|I_j| = \Delta$. Introduce a frequency cut

$$\Gamma_0 := (\log T)^2,$$

and split the spectrum into $\mathcal{L} = \{\rho : |\gamma| < \Gamma_0\}$ and $\mathcal{H} = \{\rho : |\gamma| \geq \Gamma_0\}$.

A.2. Blockwise Affine Projection and a Mean-Variance Lemma. Let P_j denote the $L^2(I_j)$ projection onto affine functions $\{a+bt\}$ and set $R_j = (\text{Id} - P_j)H_\sigma$. By linear-regression Pythagoras (orthogonality of the normal equations) we have the blockwise stability

$$(A.1) \quad \int_{I_j} |R'_j(t)|^2 dt \leq c_{\text{aff}}^{-1} \int_{I_j} |H'_\sigma(t)|^2 dt, \quad c_{\text{aff}} = \frac{1}{4},$$

with an absolute constant $c_{\text{aff}} \in (0, 1)$. Thus it suffices to bound $\sum_j \int_{I_j} |H'_\sigma|^2$.

A.3. Low Frequencies ($|\gamma| < \Gamma_0$). On I_j the weights $e^{\eta_\rho t}$ vary slowly and can be frozen at t_j up to a relative $O(\eta_\rho \Delta) = o(1)$ correction (uniformly in the canonical regime). Using $\int_{I_j} \cos^2(\gamma t) dt \asymp \Delta$ uniformly for $|\gamma| < \Gamma_0$ and the pointwise bound $\eta_\rho^2 \cos^2 + \gamma^2 \sin^2 \leq \eta_\rho^2 + \gamma^2$, we obtain

$$(A.2) \quad \int_{I_j} |H'_{\sigma, \mathcal{L}}(t)|^2 dt \ll \Delta \sum_{\rho \in \mathcal{L}} \frac{e^{2\eta_\rho t_j}}{|\rho|^{2\alpha}} (\eta_\rho^2 + \gamma^2),$$

with an absolute implied constant.

A.4. High Frequencies ($|\gamma| \geq \Gamma_0$). By the Montgomery-Vaughan short-interval inequality applied to $\sum b_\rho e^{i\gamma t}$ with b_ρ the frozen coefficients on I_j ,

$$(A.3) \quad \int_{I_j} |H'_{\sigma, \mathcal{H}}(t)|^2 dt \ll \sum_{\rho \in \mathcal{H}} \frac{e^{2\eta_\rho t_j}}{|\rho|^{2\alpha}} (\eta_\rho^2 + \gamma^2) \min\left\{\Delta, \frac{1}{\gamma^2 \Delta}\right\}.$$

Since $|\gamma| \geq \Gamma_0 \gg 1/\Delta$ in the canonical regime, the minimum equals Δ , so (A.3) has the same form as (A.2) up to constants.

A.5. Summation over Blocks and Zeros. Summing (A.2)–(A.3) over j and using

$$\sum_j \Delta e^{2\eta_\rho t_j} \ll \begin{cases} \frac{e^{2\eta_\rho T}}{\eta_\rho}, & \eta_\rho > 0, \\ T, & \eta_\rho \leq 0, \end{cases}$$

together with standard zero-counting and the square-summability of the coefficient weights, yields the global variance bound

$$(A.4) \quad \sum_j \int_{I_j} |H'_\sigma(t)|^2 dt \leq C_0(\sigma, \alpha) T \log T \log \log T,$$

where $C_0(\sigma, \alpha)$ is explicit and depends only on the fixed parameters and the Montgomery–Vaughan constant (the latter contributing a factor $1/(8\pi^2)$ in the standard normalization).

Combining (A.1) and (A.4) gives the form used in Step 10.3 of Part 2:

$$(A.5) \quad \sum_j \int_{I_j} |R'_j(t)|^2 dt \leq C'_0(\sigma, \alpha) T \log T \log \log T, \quad C'_0 = c_{\text{aff}}^{-1} C_0.$$

Remark A.1 (About lower bounds). Only the *upper* variance bound (A.4) is needed for Theorem 6.1 and for the residual control in Part 2. Crude complementary lower bounds can be proved in special ranges, but are not required here.

A.6. Numerical Verification. The constants c_{aff} , C_0 , and C'_0 have been checked in the supplementary notebooks (Part 3) by blockwise evaluation on synthetic signals and Odlyzko windows; see the repository cited in Section 12.

APPENDIX B. DERIVATIVE–VARIANCE (AFFINE POINCARÉ) CONSTANT

Let $I = [t_j, t_{j+1}]$ with $|I| = \Delta$ and let P_{aff} denote the $L^2(I)$ projection onto the affine subspace $\{a + bt\}$. For any $r \in H^1(I)$ satisfying the orthogonality conditions

$$P_{\text{aff}} r = 0 \iff \int_I r(t) dt = 0, \quad \int_I t r(t) dt = 0,$$

the sharp affine Poincaré (a.k.a. Friedrichs) inequality holds:

$$(B.1) \quad \int_I |r'(t)|^2 dt \geq \frac{4\pi^2}{\Delta^2} \int_I |r(t)|^2 dt.$$

Equivalently,

$$\int_I |r(t)|^2 dt \leq \frac{\Delta^2}{4\pi^2} \int_I |r'(t)|^2 dt.$$

Normalization used in Part 2. Writing the bound as

$$(B.2) \quad \int_I |r'(t)|^2 dt \geq \frac{c_{\text{der}}}{\Delta^2} \int_I |r(t)|^2 dt, \quad c_{\text{der}} = 4\pi^2,$$

matches the scaling adopted in equations (8.1) and (8.3).

Sharpness and proof sketch. The best constant in (B.1) is the first positive eigenvalue of $-d^2/dt^2$ on I with *natural* (Neumann) boundary conditions, restricted to functions L^2 -orthogonal to the affine subspace $\text{span}\{1, t\}$. After an affine change of variables $t \mapsto x \in [0, 1]$, the Neumann eigenfunctions are $\cos(n\pi x)$ with eigenvalues $(n\pi)^2$. Orthogonality to the constant removes $n = 0$; orthogonality to x removes $n = 1$, because $\int_0^1 x \cos(\pi x) dx = -2/\pi^2 \neq 0$ whereas $\int_0^1 x \cos(2\pi x) dx = 0$. The minimizer is therefore $\cos(2\pi x)$, giving the optimal eigenvalue $(2\pi)^2 = 4\pi^2$ on $[0, 1]$, and hence $4\pi^2/\Delta^2$ on I after rescaling to length Δ . Directly,

$$\frac{\int_0^1 (\partial_x \cos(2\pi x))^2 dx}{\int_0^1 (\cos(2\pi x))^2 dx} = \frac{2\pi^2}{1/2} = 4\pi^2.$$

Discrete grids and numerical check. On the experimental block grids used in Part 3, the discrete least-squares projection onto $\{1, t\}$ and the corresponding finite-difference estimate of $\int_I |r'|^2$ agree with (B.1) to high precision: the supplementary routine `verify_c_der` reports an empirical minimum of 39.483 for the Rayleigh quotient over random trigonometric residuals, matching $c_{\text{der}} = 4\pi^2 \approx 39.478$ and confirming the Δ^{-2} scaling.

Remark. Inequality (B.1) is uniform in the block position t_j and depends only on the block length Δ ; no assumptions on r beyond $r \in H^1(I)$ and $P_{\text{aff}} r = 0$ are required.

APPENDIX C. CONSTANT-EXTRACTION METHODOLOGY (OPTIONAL)

This appendix documents how numerical constants and slopes were extracted from the experiments in Part 3. All scripts and CSVs (`bias_summary.csv`, `bias_blocks.csv`, `variance_table.csv`) are included in the repo and reproduce Figures 1–4 and the tables cited there.

Common settings. Unless otherwise stated, blocks have $\Delta = c_\Delta / \log T$ with $c_\Delta = 1$, and the penalty is $\lambda = c_\lambda (\log T)^{-2}$ with $c_\lambda \in \{1/4, 1, 4\}$. Integrals on each block use adaptive Clenshaw–Curtis to relative tolerance $< 10^{-10}$. All runs were performed in R 4.3.1 (double precision).

Data windows. Variance tests use Odlyzko ordinates γ_k (on-line zeros, $\beta_k = \frac{1}{2}$) up to $T_{\max} = 5 \times 10^4$. Bias tests augment the on-line set by a synthetic off-line zero at $\beta = \sigma + \eta$, with $\eta \in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}, 10^{-5}\}$.

1) Derivative constant c_{der} . We use the affine Poincaré inequality on a block I of length Δ :

$$\int_I |r'(t)|^2 dt \geq \frac{4\pi^2}{\Delta^2} \int_I |r(t)|^2 dt.$$

Thus $c_{\text{der}} = 4\pi^2$ in the normalization of Part 2 (see Appendix B). Numerically, we verify sharpness by projection onto the affine subspace on each block and computing $\|r'\|_{L^2(I)}^2 / \|r\|_{L^2(I)}^2$ over 10^4 randomly phased test signals $\sum_{m=1}^M a_m \cos(\omega_m t + \phi_m)$ with $M \in [1, 8]$, $\omega_m \Delta \in [0.1, 10]$. The empirical minimum equals $4\pi^2 / \Delta^2$ (for $\Delta = 1$, 39.483 versus $4\pi^2 = 39.478$).

2) Variance constants C_0, C'_0 . We bound $\sum_j \int_{I_j} |H'_\sigma(t)|^2 dt \leq C_0 T \log T \log \log T$ as in Appendix A. From `variance_table.csv` (columns $F_\lambda, T \log T \log \log T$) we regress $Y = \sum_j \int_{I_j} |H'_\sigma|^2$ on $X = T \log T \log \log T$ *through the origin* to obtain $\hat{C}_0$. Reported intervals are nonparametric percentile bootstrap ($B = 2000$ resamples over T), typically $\hat{C}_0 \pm 0.002$. The affine residual bound (Equation (A.1)) yields $C'_0 = c_{\text{aff}}^{-1} C_0$ with $c_{\text{aff}} = \frac{1}{4}$.

3) Slope calibration (bias regime). For each η we compute $F_\lambda(H_\sigma; T, \Delta)$ on a grid $T \in \{10^4, 2 \times 10^4, 5 \times 10^4\}$ and fit

$$\log F_\lambda(T) = sT + b + \varepsilon$$

using Huber regression (tuning $k = 1.345$) to reduce leverage from the smallest T . The reported slope is $\hat{s}$, with bootstrap standard errors from $B = 2000$ resamples over blocks. Relative errors against the theoretical 2η are recorded in

REFERENCES

- [1] A. M. Odlyzko, *Tables of zeros of the Riemann zeta function*, https://www-odlyzko.dtc.umn.edu/zeta_tables/.
- [2] H. L. Montgomery and R. C. Vaughan, *The large sieve*, *Mathematika* **20** (1973), 119–134.
- [3] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, *Selecta Math. (N.S.)* **5** (1999), 29–106.
- [4] M. V. Berry and J. P. Keating, *$H = xp$ and the Riemann zeros*, in *Supersymmetry and Trace Formulae: Chaos and Disorder*, Kluwer, 1999, pp. 355–367.
- [5] N. L. Balazs and A. Vörös, *Chaos on the pseudosphere*, *Phys. Rep.* **143** (1986), 109–240.
- [6] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986.
- [7] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, Amer. Math. Soc. Colloq. Publ. **53**, 2004.
- [8] A. E. Ingham, *The Distribution of Prime Numbers*, Cambridge University Press, 1932.
- [9] J. Pintz, *On the remainder term of the prime number formula and the zeros of the Riemann zeta-function*, in *Number Theory (Noordwijkerhout 1983)*, Lecture Notes in Math. **1068**, Springer, 1984, pp. 186–197.
- [10] L. Báez-Duarte, *A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis*, *Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl.* **14** (2003), 5–11.
- [11] A. Beurling, *A closure problem related to the Riemann zeta-function*, *Proc. Natl. Acad. Sci. USA* **41** (1955), 312–314.
- [12] X.-J. Li, *The positivity of a sequence of numbers and the Riemann hypothesis*, *J. Number Theory* **65** (1997), 325–333.
- [13] E. Bombieri and J. C. Lagarias, *Complements to Li’s criterion for the Riemann hypothesis*, *J. Number Theory* **77** (1999), 274–287.
- [14] J. C. Lagarias, *An elementary problem equivalent to the Riemann hypothesis*, *Amer. Math. Monthly* **109** (2002), 534–543.
- [15] A. Speiser, *Geometrisches zur Riemannschen Zetafunktion*, *Math. Ann.* **110** (1934), 514–521.
- [16] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in *Analytic Number Theory*, Proc. Sympos. Pure Math. **24**, Amer. Math. Soc., 1973, pp. 181–193.
- [17] Z. Rudnick and P. Sarnak, *Zeros of principal L-functions and random matrix theory*, *Duke Math. J.* **81** (1996), 269–322.
- [18] J. B. Conrey, *The Riemann hypothesis*, *Notices Amer. Math. Soc.* **50** (2003), 341–353.
- [19] P. Sarnak, *Asymptotic behavior of periodic orbits of the horocycle flow and Eisenstein series*, *Comm. Pure Appl. Math.* **34** (1981), 719–739.

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